Title: Anaya Scott Date: / / M T W T F S S

Quiz 3

$$e^{\epsilon} - e^{-\epsilon}$$

$$e^{\epsilon} = 1 + \epsilon + \frac{\epsilon^2}{2!} + \frac{\epsilon^3}{3!} + \frac{\epsilon^4}{4!} + \dots$$

$$e^{-\epsilon} = 1 - \epsilon + \frac{\epsilon^2}{2!} - \frac{\epsilon^3}{3!} + \frac{\epsilon^4}{4!} - \dots$$

$$\left(\cancel{1} + \epsilon + \cancel{\frac{\epsilon^2}{2!}} + \frac{\epsilon^3}{3!} + \cancel{\frac{\epsilon^4}{4!}} + \frac{\epsilon^5}{5!}\right) - \left(\cancel{1} - \epsilon + \cancel{\frac{\epsilon^2}{2!}} - \frac{\epsilon^3}{3!} + \cancel{\frac{\epsilon^4}{4!}} - \frac{\epsilon^5}{5!}\right)$$

$$2\epsilon + \frac{2\epsilon^3}{3!} + \frac{2\epsilon^5}{5!} + \dots$$

$$\downarrow$$

Since we only go up to the second order

$$e^{\epsilon} - e^{-\epsilon} \approx \boxed{2\epsilon}$$